

RAG

RETRIEVAL-AUGMENTED GENERATION STRATEGIES TO
PERSONALISE INTERACTIONS IN AN LLM-BASED
CHATBOT

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Problem & Research Question

- P** LLMs are fluent but ungrounded. RAG fixes factual grounding but most RAG research targets generic retrieval not personalising support to an individual user.
- R1** Which RAG strategy performs best overall for e-commerce support?
- R2** Does Contextual RAG improve personalisation over generic retrieval?
- R3** Does Agentic RAG trade latency for completeness?

Five RAG Strategies Compared

Strategy	Retrieval	Personalisation	Trade-off
Baseline	Dense vector search	Persona reranked with KB	Simple reference point
Multi-Query	3 LLM-paraphrased queries, merged	Persona reranked with KB	Wider recall, more LLM calls
Contextual	Original query + separate persona lookup	Persona chunks + customer name to LLM	Flagship personalisation strategy
Hybrid	BM25 + dense, fused	Persona reranked with KB	Built for exact-match (order ID, product ID)
Agentic	ReAct agent picks tools dynamically	GetCustomerProfile() tool	Most flexible, ~5s latency

Evaluation Setup

RAGAS (AUTOMATED):

20 GENERAL QUERIES + 8 PERSONALISED QUERIES - SCORED ON FAITHFULNESS,
RELEVANCY, CONTEXT PRECISION/RECALL

HUMAN EVALUATION:

9 PARTICIPANTS EVALUATING ON SYSTEM A-E - SCORED ON ACCURACY, CLARITY,
PERSONALISATION, TRUST (1-10)

Results

AUTOMATED METRICS (RAGAS, 20 GENERAL QUERIES):

- AGENTIC RAG PERFORMED THE BEST ON FAITHFULNESS (0.933) AND CONTEXT PRECISION (0.958) BUT IS BY FAR THE SLOWEST (5.638S)
- HYBRID PERFORMED THE WORST ON FAITHFULNESS (0.825) AND ANSWER RELEVANCY (0.864) BUT IT WAS THE FASTEST (1.871S)

Results

AUTOMATED METRICS (RAGAS, 8 PERSONALISED QUERIES):

- CONTEXTUAL RAG HIGHEST ON ANSWER RELEVANCY (0.69), FAITHFULNESS (0.854) AND FAST WAS THE FASTEST (2.172)
- CONTEXT PRECISION/RECALL NEAR-IDENTICAL ACROSS ALL FIVE BECAUSE PERSONA FILE IS SMALL

Results

HUMAN EVALUATION

- CONTEXTUAL RAG: 8.33/10 — HIGHEST, AND MOST-VOTED "MOST PERSONALISED"
- AGENTIC RAG: SECOND (7.81/10) — MOST-VOTED "MOST PREFERRED" AND "MOST HELPFUL"
- BASELINE / MULTI-QUERY / HYBRID: 6.4–7.2/10

Hypotheses & Limitations

- L1** Human-eval sample is small (n=9)
- L2** Contextual RAG doesn't rerank persona & KB chunks together like the other four architectural
- H1** Contextual best for personalisation - Supported by human evaluation
- H2** Hybrid beats Baseline on product/order queries - Untestable: results weren't broken down by query category
- H3** Agentic trades latency for completeness - Supported: highest automated scores, ~3x latency, most-preferred overall

Conclusion

Strategies that explicitly bring in customer-specific data , through context injection (Contextual) or dynamic tool lookup (Agentic), are clearly preferred by real users over generic retrieval, even where automated metrics doesn't show one clean winner.

Future work

- **Break down RAGAS results by query category to properly test H2**
- **Fix Contextual RAG's rerank and re-evaluate**

Any Questions?

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